NUS HIGH SCHOOL of Math & Science

National Mathematical Olympiad of Singapore

SPECIAL ROUND

Date: 1 AUGUST 2023

Time Given: 1 hour 30 minutes

Instructions to Candidates

1. Do not open the booklet until you are told to do so.
2. Answer as many questions as you can.
3. Write your answers in the answer sheet provided and shade the appropriate bubbles below your answers.
4. No steps are needed to justify your answers.
5. Questions 1 – 5 are worth 3 marks each.
6. Questions 6 – 10 are worth 4 marks each.
7. Questions 11 – 20 are worth 5 marks each.
8. No marks will be deducted for wrong answers.
9. Unanswered questions will not get any marks.
10. No calculators or mathematical instruments are allowed.

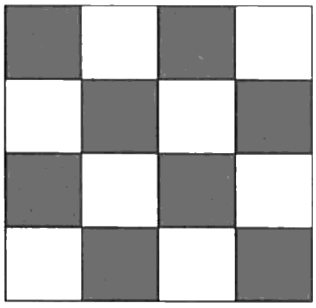
Questions 1 to 5 are worth 3 marks each.

1. A Mathematics Test was conducted to a class of Primary 5 students. If the highest score of 96 is removed from the class, the average score decreases to 75. If the lowest score of 42 is removed from the class, the average score increases to 78. How many students are there in the class?

2. The notation $n!$ means $n\times(n-1)\times(n-2)\times\cdots\times 1$, where n is a non-negative whole number. For example, $4!=4\times3\times2\times1=24$. Find the last two digits of $1!+4!+7!+10!+\cdots+2023!$. Express your answer as a two-digit number. For example, if you think the sum above is 12405, write "05".

3. In a group of adults, 40% of those wearing glasses are women and 35% of women wear glasses. If 70% of the group are men, what is the minimum number of people in the group?

4. The squares of a 4-by-4 grid are coloured grey or white in the pattern shown below.

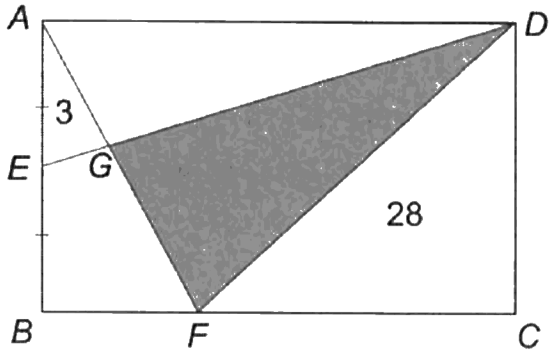


In a move, you may choose any row or column and reverse the colours of all the squares in that row or column. In other words, a white square becomes a grey square whereas a grey square becomes a white square. You may repeat these moves for any number of times. What is the smallest number of moves to change the above grid pattern into a grid of all white squares?

5. How many zeros does the number $1\times2\times3\times\cdots\times198\times199\times200$ end with?

Questions 6 to 10 are worth 4 marks each.

6. In the rectangle $ABCD$ shown below, E is the mid-point of AB, and F is a point on BC. The lines AF and DE intersect at the point G. Given that the area of triangles AEG and CDF are 3 cm^2 and 28 cm^2 respectively, find the area of triangle DFG in cm^2.

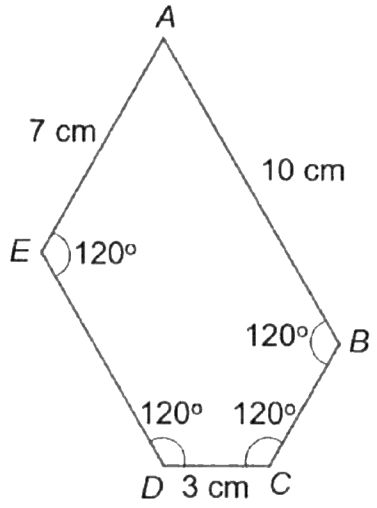


7. Alicia and Benjamin are planting trees in the garden at the same rate. Alicia starts planting when Benjamin completed $\frac{3}{7}$ of his task. When Benjamin finished his entire task, Alicia has completed $\frac{7}{11}$ of her own task. What is the minimum number of trees they have planted when both of them have completed their tasks?

8. How many four-digit numbers $\overline{abcd}$ satisfy the property that $\overline{abcd} - \overline{bcda} = 2007$? Here, we allow b to be 0.

9. A jug is initially filled with liquid consisting of 28% water and 72% oil. After 50 ml of water is poured out of the jug and 100 ml of oil is poured into the jug, the liquid in the jug now consists of 20% water and 80% oil. Find the initial amount of water in the jug (in ml).

10. The diagram below shows a pentagon $ABCDE$ such that $AB = 10$ cm, $AE = 7$ cm, $CD = 3$ cm and $\angle ABC = \angle BCD = \angle CDE = \angle DEA = 120^\circ$.

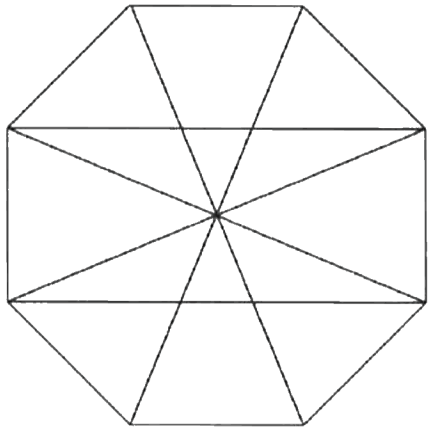


Find the perimeter of the pentagon (in cm).

Questions 11 to 20 are worth 5 marks each.

11. Find the value of $9900\times\left(1-\frac{4}{3^2}\right)\times\left(1-\frac{4}{4^2}\right)\times\left(1-\frac{4}{5^2}\right)\times\cdots\times\left(1-\frac{4}{100^2}\right)$.

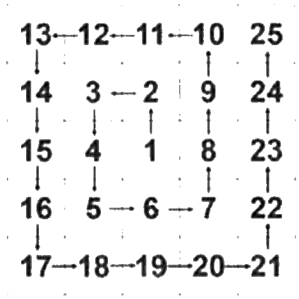
12. How many triangles are there in the diagram below?



13. Alice left her house and was on her way to meet her mother who was returning home from work. They both started their journey at the same time and travelled along the same straight pathway. Alice walked at 47 m/min while her mother walked at 51 m/min. They met at 82 m from the midpoint of the entire journey. How far, in metres, is Alice's mum's workplace from their house?

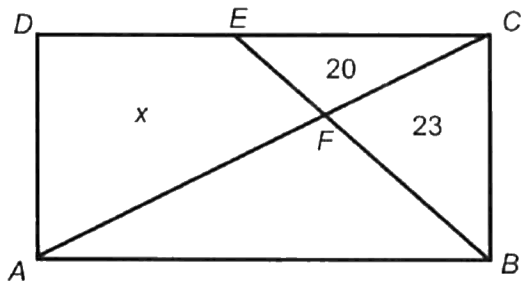
14. A whole number is considered "lucky" if its digits can be divided into two groups such that the sum of the digits in each group is the same. For example, 34175 is lucky because $3+7=4+1+5$. Find the sum of all lucky numbers less than 200 which are also prime numbers.

15. Square tiles (each labelled with a whole number) are arranged in increasing order of the number on the tiles and in an anti-clockwise fashion as shown below. This pattern continues for numbers beyond 25 as well.



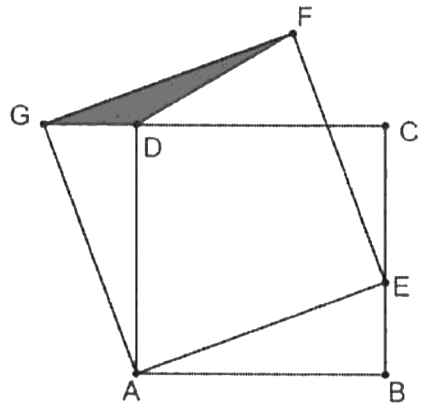
The number "southwest" of a number x is defined to be the number on the tile that is one tile south and one tile west of the tile with number x, if such a tile exists. For example, the number southwest of 2 is 4, and the number southwest of 22 is 20. What is the number southwest of 2023?

16. As shown in the figure below, $ABCD$ is a rectangle. Point E lies on the side CD and point F is the intersection point of line segments EB and AC. The areas of triangles CEF and BCF are known to be 20 cm^2 and 23 cm^2 respectively. If the area of quadrilateral $ADEF$ is x cm^2, find the value of $20x$.



17. All the factors of a whole number N (except 1 and N) are written down one after another. It turns out that the largest factor is 105 times as big as the smallest factor. Find the sum of all possible values of N.

18. As shown below, *ABCD* and *AEFG* are squares. The areas of *ABCD* and *AEFG* are 1729 cm^2 and 2023 cm^2 respectively. Find the shaded area, in cm^2, of *DFG*.



19. A 4-by-4 magic square is created with the following properties:

- It uses the numbers 1 to 16 with no repeats.
- Numbers in each row add up to *N*.
- Numbers in each column add up to *N*.
- Numbers in long diagonals in the square add up to *N*.
- Numbers in the 4 corners add up to *N*.
- Numbers in every 2 by 2 squares add up to *N*.

1			
12			
	4		10

A partially-completed square is drawn above. Complete the magic square and find the product of the numbers in the 3 shaded squares.

20. In this sequence of whole numbers, each number (starting from b) is the average of the two numbers before it.

$$1099,\ a,\ b,\ c,\ d,\ e,\ 2023$$

Find the value of c.

- THE END -

SPECIAL ROUND – FINAL ANSWER

1.	19	**11.**	400
2.	65	**12.**	28
3.	1717	**13.**	4018
4.	4	**14.**	210
5.	49	**15.**	1847
6.	24	**16.**	589
7.	93	**17.**	1365
8.	7	**18.**	123
9.	279	**19.**	462
10.	31	**20.**	2155

Remark: To access the full solutions for Special Round questions, scan the following QR code or refer to NUS High School NMOS website at http://nush.link/nmos